StreamVerseX – Real-Time OTT Analytics Platform

1. Objectives

The objective of StreamVerseX is to build an end-to-end OTT analytics platform capable of processing, storing, analyzing, and visualizing both historical and real-time streaming data. The platform helps stakeholders understand user engagement, content performance, subscriber behavior, recommendation effectiveness, and platform reliability through interactive dashboards and automated analytics pipelines.

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2. Problem Statement

Modern OTT platforms generate millions of events such as video views, watch history, subscriptions, recommendations, and user interactions. Analyzing this large volume of data in real time while also maintaining historical analytics is challenging.

Organizations need a centralized system that can:

- Process large-scale streaming data
- Track user engagement
- Monitor platform health
- Measure content performance
- Analyze subscriber trends
- Generate business insights in real time

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3. Goal

The goal of StreamVerseX is to create a scalable OTT analytics ecosystem that combines:

- Real-Time Event Streaming
- Automated ETL Pipelines
- Data Warehousing
- Business Intelligence
- Interactive Dashboards

to support data-driven decision making.

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4. Data Generation

The project simulates OTT platform activities through:

User Events

- Video Plays

- Watch History
- Likes
- Content Completion
- Recommendations
- Search Events

Kafka Streaming Events

Real-time events are generated using Kafka Producer:


```
{
  "user_id": 43880,
  "video_id": "VID2286",
  "event_type": "LIKE",
  "watch_time": 1623
}
```


These events continuously flow through Kafka topics and are consumed into PostgreSQL.

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5. Data Storage Layer

Database

PostgreSQL

Raw Data Tables

- streaming_sessions
- user_watch_history
- subscription_transactions
- cdn_buffering_logs
- content_metadata
- app_clickstream_events
- search_recommendation_logs

Real-Time Table

- realtime_events

This layer acts as the primary storage system for OTT platform data.

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6. Analytical Layer (Data Warehouse)

A Star Schema architecture was designed for analytical processing.

Dimension Tables

- dim_user
- dim_content
- dim_device
- dim_region
- dim_date

Fact Tables

- fact_streaming_sessions_dw
- fact_watch_history_dw
- fact_subscriptions_dw
- fact_clickstream_dw
- fact_recommendations_dw

Additional Components

- Views
- Materialized Views
- Stored Procedures
- Triggers
- Alert Tables

This layer transforms raw data into analytics-ready datasets.

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7. Architecture

OTT User Activity
↓
Kafka Producer
↓
Apache Kafka
↓
Kafka Consumer
↓
PostgreSQL
↓
Apache Airflow ETL
↓
Data Warehouse (Star Schema)
↓
FastAPI APIs
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Power BI Dashboards

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React Analytics Portal

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8. Visualization Layer

Analytics are visualized through Power BI and React dashboards.

Dashboard 1 – OTT Command Center

Provides executive-level KPIs:

- Total Users
- Revenue
- Streams
- Recommendations
- Watch Time

Dashboard 2 – Content Intelligence

Analyzes:

- Genre Popularity
- Watch Time
- Completion Rate
- Content Distribution

Dashboard 3 – Subscriber Intelligence

Tracks:

- Revenue by Plan
- Subscribers
- Renewals
- Successful Payments

Dashboard 4 – Platform Reliability

Monitors:

- Crash Rate
- Platform Events
- Device Performance
- Streaming Quality

Dashboard 5 – Real-Time Analytics

Powered by Kafka:

- Realtime Events
- Active Users
- Watch Time
- Video Activity

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9. Key Insights

Content Insights

- Action and Drama categories generated the highest engagement.
- Watch time distribution varies across genres.
- Completion rates remained close to 50% across categories.

Subscriber Insights

- Premium plans generated higher revenue.
- Renewal rates indicate strong subscriber retention.
- Revenue contribution differs across subscription plans.

Platform Insights

- Certain screens experienced higher crash rates.
- Platform reliability can be monitored proactively using trigger-generated alerts.

Real-Time Insights

- Kafka continuously streams user events.
- Active user and watch activity metrics can be monitored dynamically.

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10. Business Insights

- High-performing content categories can be prioritized for future investments.
- Subscriber analytics help optimize pricing and retention strategies.
- Recommendation effectiveness can improve user engagement.
- Reliability monitoring reduces user churn caused by technical issues.
- Real-time event tracking enables faster operational decision-making.

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11. Conclusion

StreamVerseX successfully integrates Data Engineering, Real-Time Streaming, Data Warehousing, Business Intelligence, and Full-Stack Development into a single analytics platform.

The solution demonstrates how modern OTT platforms can leverage Kafka, Airflow, PostgreSQL, FastAPI, React, Docker, and Power BI to transform raw streaming data into actionable business intelligence and real-time operational insights.

Technologies Used

- PostgreSQL
- Apache Kafka
- Apache Airflow
- FastAPI
- React.js
- Power BI
- Docker
- GitHub